

Exact algebraic structure of hyperbolic hybrid tilings

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Abstract

A *hybrid* system in the hyperbolic plane is a collection of vertex combinations of regular polygons that close at a common edge length, so that they may occur in a single tiling. Catalogues of such systems have so far been built numerically: the shared edge lengths are computed by iteration to some fifty decimal places, and the coincidences are believed but not proved. We show that these edge lengths are algebraic numbers, that a defining polynomial for each is computable by elimination, and that the question of whether two vertex combinations share an edge is decidable exactly. Applying the procedure we give closed forms for eight catalogued edges, among them $\cosh s = \varphi$, $\cosh s = (3 + 8\sqrt{2})/7$ and $s = \log 3$, and we prove exact every identity in those systems. The apparatus is insensitive to the sign of the curvature, so it treats spherical and Euclidean systems at the same time. For the system at $\cosh s = \varphi$ we then give a complete additive description: all twelve polygons occurring at integer multiples of the base edge have interior angles lying in a rank-three lattice spanned by π , $\arccos(1/\varphi)$ and $\arccos(-1/\varphi^2)$, so that enumerating the vertex combinations reduces to three linear Diophantine equations. Finally we classify the systems in which an equilateral triangle at r times the base edge has r times the area, obtaining $\cosh s = 1 + \sqrt[3]{2} + \sqrt[3]{4}$, $(5 + \sqrt{33})/4$, $(3 + \sqrt{33})/4$ and φ for $r = 2, 3, 4, 7$, and showing that these are the only values of r for which further polygons occur.

1 Introduction

Let $\mathbb{H}^2$ denote the hyperbolic plane of curvature -1 . A regular n -gon with side length s exists in $\mathbb{H}^2$ for every $n \geq 3$ and every $s > 0$, and its interior angle $\theta_n(s)$ decreases from the Euclidean value $(1 - 2/n)\pi$ towards 0 as s grows. The apeirogon, which we write $n = \infty$, is the limiting case: a regular polygon with infinitely many sides of length s inscribed in a horocycle.

Because the angle depends on the side, polygons of different shapes can be made to fit around a vertex by choosing the side length appropriately. This is the mechanism behind what the first author's working notes call *hybrid* systems: a set of vertex combinations that all close at one and the same edge length. Such a set is the raw material of a tiling in which several distinct regular polygons occur, all with a common side.

The catalogues of these systems have been assembled experimentally. For each candidate vertex combination one solves a transcendental equation numerically for the edge, records the result to high precision, and groups combinations whose recorded edges agree. Agreement to fifty decimal places is persuasive, and a mismatch at that precision would be strange, but it is not a proof. Recording this as an open problem, those notes ask for the identities to be established exactly.

That is the first thing we do here. The key observation is elementary: the closure condition at a vertex, which as stated is a transcendental equation in the edge, becomes a *multiplicative* relation among algebraic numbers on the unit circle once one passes to half-angles (Theorem 3.1). From this the edge is seen to be algebraic, an explicit defining

polynomial is computable by resultants, and the question of whether two combinations share an edge becomes a greatest common divisor computation (Theorem 3.4). Section 4 carries this out for eight systems of the catalogue.

The remainder of the paper studies one system in detail, the one at $\cosh s = \varphi$ where $\varphi = (1 + \sqrt{5})/2$, together with the polygons that occur at integer multiples of its base edge. Section 5 shows that all twelve such polygons have interior angles in a lattice of rank three, which reduces the enumeration of their vertex combinations to linear algebra over $\mathbb{Z}$. Section 6 isolates the geometric construction behind two of the coincidences and determines exactly when it applies. Section 7 classifies a one-parameter family of systems singled out by an area condition.

Throughout we distinguish carefully between statements we prove and statements established by computation over a bounded search range; the latter are labelled as computational observations.

2 Preliminaries

Lemma 2.1. *Let P be a regular n -gon in $\mathbb{H}^2$ with side length s and interior angle θ , where $3 \leq n \leq \infty$. Write $c = \cosh(s/2)$ and $c_n = \cos(\pi/n)$, with $c_\infty = 1$. Then*

$$\sin \frac{\theta}{2} = \frac{c_n}{c}, \quad \cos \frac{\theta}{2} = \frac{\sqrt{c^2 - c_n^2}}{c}. \quad (1)$$

The polygon exists precisely when $c > c_n$, and then $\theta \in (0, \pi)$.

Proof. Join the centre of P to a vertex and to the midpoint of an adjacent side. This produces a right triangle with legs meeting at the centre in the angle π/n , at the vertex in the angle $\theta/2$, and with the side opposite the central angle equal to $s/2$. The hyperbolic right-triangle relation $\cosh a = \cos A / \sin B$, where a is the side opposite A and B is the third angle, gives $\cosh(s/2) = \cos(\pi/n) / \sin(\theta/2)$, which is the first identity. Since $\theta \in (0, \pi)$ we have $\theta/2 \in (0, \pi/2)$, so the cosine is positive and the second identity follows. The apeirogon is the limit $n \rightarrow \infty$, for which $c_n \rightarrow 1$. $\square$

Remark 2.2 (Curvature is a sign, not a case). Nothing below uses $c > 1$. In all three geometries of constant curvature the same relation holds,

$$c = \frac{\cos(\pi/n)}{\sin(\theta/2)}, \quad c = \begin{cases} \cos(s/2) & \text{on the sphere,} \\ 1 & \text{in the Euclidean plane,} \\ \cosh(s/2) & \text{in } \mathbb{H}^2, \end{cases}$$

the spherical case being the substitution $s \mapsto is$. Thus c alone carries the geometry: a solution of the polynomial P_m of Theorem 3.2 with $c < 1$ is a spherical vertex, $c = 1$ a Euclidean one, and $c > 1$ hyperbolic. For instance the cube $\{4, 3\}$ has $c = \cos(\pi/4) / \sin(\pi/3) = \sqrt{2/3}$, and its edge is $2 \arccos c = 70.5288 \dots^\circ$. We phrase everything for $\mathbb{H}^2$ because that is where the systems are plentiful, but every statement that does not mention $\cosh$ applies verbatim to the other two geometries.

Remark 2.3 (Shortchord, and cyclic polygons). The constant $c_n = \cos(\pi/n)$ is half the *shortchord* of a Euclidean regular n -gon of side 1, that is half its shortest diagonal. With that reading, (2) says $\sum_i 2 \arcsin(a_i/(2R)) = 2\pi$ with $a_i = 2c_{n_i}$ and $R = c$, which is exactly the condition that a Euclidean polygon with sides a_i be inscribed in a circle of radius R containing its centre. So P_m is also a defining polynomial for the circumradius of a cyclic polygon with prescribed sides. That problem is not open: Robbins [9] obtained the analogous

area polynomial in 1994 and Pech [8] the circumradius polynomial in 2006, both of degree 7 in the pentagon and heptagon cases, with later work by Varfolomeev and others; explicit general formulas beyond small n remain hard. The elimination of Theorem 3.2 is an independent route to those polynomials, and conversely their degree bounds are the natural place to look for the height bound requested in Section 8.

For fixed n the map $c \mapsto \theta_n(c) = 2 \arcsin(c_n/c)$ is strictly decreasing on (c_n, ∞) , with $\theta_n(c) \rightarrow \pi - 2\pi/n$ as $c \downarrow c_n$ and $\theta_n(c) \rightarrow 0$ as $c \rightarrow \infty$.

Definition 2.4 (Vertex combination and edge function). A *vertex combination* is a function m from $\{3, 4, 5, \dots\} \cup \{\infty\}$ to $\mathbb{Z}_{\geq 0}$ with finite support, written multiplicatively as $(n_1^{m_1} n_2^{m_2} \dots)$. Its *degree* is $M = \sum_n m_n$. We say m *closes* at c if

$$\sum_n m_n \theta_n(c) = 2\pi. \quad (2)$$

Lemma 2.5. *A vertex combination closes at no more than one value of c .*

Proof. On the common domain $c > \max\{c_n : m_n > 0\}$ each θ_n is strictly decreasing, hence so is the finite positive combination $\sum_n m_n \theta_n$. A strictly monotone function attains the value 2π at most once. $\square$

When the value exists we write $e(m) = c$ for it, and call it the *edge function* of m ; we shall also refer to $s = 2 \operatorname{arcosh} c$ as the edge. Two vertex combinations m, m' with $e(m) = e(m')$ are said to satisfy a *hybrid identity*; a maximal set of combinations pairwise satisfying one is a *hybrid system*. Only combinations in a common system can occur in a single tiling by regular polygons of one side length.

3 The closure condition is a multiplicative relation

Fix a vertex combination m with support S and degree M . For $n \in S$ introduce the auxiliary quantity $u_n = \sqrt{c^2 - c_n^2}$, so that by Lemma 2.1

$$z_n = e^{i\theta_n/2} = \frac{u_n + i c_n}{c}.$$

Theorem 3.1. *With the notation above, m closes at c if and only if*

$$\prod_{n \in S} (u_n + i c_n)^{m_n} = -c^M, \quad (3)$$

and (3) implies, more weakly, $\sum_n m_n \theta_n \equiv 2\pi \pmod{4\pi}$.

Proof. Condition (2) is equivalent to $\sum_n m_n \theta_n/2 = \pi$, hence to $\prod_n z_n^{m_n} = e^{i\pi} = -1$. Multiplying by c^M gives (3). Conversely (3) gives $\prod_n z_n^{m_n} = -1$, that is $\sum_n m_n \theta_n/2 \equiv \pi \pmod{2\pi}$, which is the stated congruence. $\square$

The point of (3) is that its two sides are polynomials in c , the u_n , and the algebraic numbers c_n , whereas (2) is transcendental. Separating real and imaginary parts, the imaginary part of the left side of (3) must vanish; that is a single polynomial condition.

Theorem 3.2. *Let m be a vertex combination that closes. Then $e(m)$ is an algebraic number, and a nonzero polynomial $P_m \in \mathbb{Q}(\{c_n\}_{n \in S})[X]$ with $P_m(e(m)) = 0$ is computable by the following procedure.*

1. Expand $W = \prod_{n \in S} (U_n + i c_n)^{m_n}$ as a polynomial in independent indeterminates U_n .

2. Reduce every U_n^2 to $X^2 - c_n^2$, so that each U_n occurs to degree at most one.
3. Let $E = \text{Im } W$, a polynomial in X and the U_n .
4. Successively eliminate the U_n by $E \mapsto \text{Res}_{U_n}(E, U_n^2 - (X^2 - c_n^2))$.

The result P_m is a polynomial in X alone, and it is even in X .

Proof. By Theorem 3.1 the number $c = e(m)$, together with the positive square roots $u_n = \sqrt{c^2 - c_n^2}$, satisfies $\text{Im } W = 0$ after substituting $X = c$, $U_n = u_n$. Steps 2 and 4 are the standard elimination of the u_n : the resultant of E with the minimal polynomial of u_n over $\mathbb{Q}(\{c_j\}, X)$ lies in the ideal generated by them, so it vanishes at $X = c$, and it no longer involves U_n . After all eliminations $P_m(c) = 0$. Nonvanishing of P_m as a polynomial is checked in the course of the computation. Since θ_n depends on c only through c^2 in (1) after the substitution $u_n^2 = c^2 - c_n^2$, the surviving polynomial is even in X ; we write $P_m(X) = F_m(X^2)$. $\square$

Remark 3.3. The root set of P_m is in general larger than $\{e(m)\}$. Two sources of spurious roots are visible in the proof: the congruence in Theorem 3.1 admits $\sum m_n \theta_n \in \{2\pi, 6\pi, \dots\}$, and the elimination step forgets the sign conditions $u_n > 0$. This does not affect what follows, because we always identify the relevant root by numerical isolation.

Theorem 3.4 (Decision procedure). *Let m, m' be vertex combinations that close, with $P_m, P_{m'}$ as above, and let $G = \text{gcd}(P_m, P_{m'})$, computed exactly. Suppose $I \subset \mathbb{R}$ is an interval such that*

1. P_m has exactly one root in I , and $P_{m'}$ has exactly one root in I ;
2. $e(m) \in I$ and $e(m') \in I$;
3. G has a root in I .

Then $e(m) = e(m')$. If instead G has no root in I , then $e(m) \neq e(m')$.

Proof. Let $\alpha \in I$ be a root of G . Then α is a root of P_m , and $e(m)$ is a root of P_m lying in I ; by hypothesis (1) they coincide, so $\alpha = e(m)$. The same argument applied to $P_{m'}$ gives $\alpha = e(m')$. Hence $e(m) = e(m')$. If G has no root in I then $e(m)$, being in I , is not a common root, so $e(m) \neq e(m')$. $\square$

All three hypotheses are effectively checkable: root counting in an interval is Sturm's theorem over $\mathbb{Q}$, or root isolation over the coefficient field in general, and $e(m)$ is located to any desired precision by Lemma 2.5 together with bisection. The procedure therefore decides hybrid identities. This settles, in principle, the question of whether the numerically observed coincidences are exact; the remaining difficulty is computational, and we return to it in Section 8.

4 Exact edges for eight catalogued systems

We have run the procedure of Theorems 3.2 and 3.4 on part of the catalogue. Table 1 records the outcome. In each case the polynomials F_m were computed symbolically, the greatest common divisor was taken exactly, and the resulting value of $\cosh s = 2c^2 - 1$ was solved in closed form, where the degree permits, and then verified against every member combination of the system to eighty decimal digits. The quartic at 1.1555 is irreducible over $\mathbb{Q}$; we give its minimal polynomial instead of a radical expression.

s (as catalogued)	vertex combinations	$\cosh s$ exactly
0.6196...	$(3^2.5.\infty), (3.5^3)$	$\frac{2 + 5\sqrt{5}}{11}$
1.0612...	$(3^4.4^2), (3^2.4.5^2), (3.4.10.20), (5^4)$	$\varphi = \frac{1 + \sqrt{5}}{2}$
1.0986...	$(4^2.5.\infty), (\infty^3)$	$\frac{5}{3}$, so $s = \log 3$
1.1555...	$(3.5.12^2), (4.5^2.12)$	root of $191y^4 - 424y^3 + 10y^2 + 328y - 121$
1.2537...	$(4^5), (4^2.10.30)$	$\frac{5 + 2\sqrt{5}}{5} = 1 + \frac{2}{\sqrt{5}}$
1.3424...	$(3^3.\infty^2), (3.8.24.\infty)$	$\frac{3 + 8\sqrt{2}}{7}$
1.6628...	$(3^4.\infty^2), (3^2.12^2.\infty), (12^4)$	$1 + \sqrt{3}$
1.7627...	$(4^6), (4^3.\infty^2), (\infty^4)$	3, so $s = \log(3 + 2\sqrt{2})$

Table 1: Exact edge lengths. Every hybrid identity in these eight systems is a theorem, by Theorem 3.4.

Example 4.1. Take $(3^4.4^2)$, four triangles and two squares. Here $c_3 = \frac{1}{2}$, $c_4 = \frac{\sqrt{2}}{2}$, $M = 6$, and the elimination of Theorem 3.2 yields $P(X) = 4X^4 - 6X^2 + 1$, so $c^2 = (3 + \sqrt{5})/4$ and

$$\cosh s = 2c^2 - 1 = \frac{1 + \sqrt{5}}{2} = \varphi.$$

The same value is returned by (5^4) , by $(3^2.4.5^2)$ and by $(3.4.10.20)$, and in each case the greatest common divisor of the two polynomials is of positive degree, so the identity is exact.

Remark 4.2. Two of the eight edges satisfy $\cosh s \in \mathbb{Q}$, whence e^s is an algebraic unit and s is the logarithm of one: $s = \log 3$ and $s = \log(3 + 2\sqrt{2})$. We note in passing that s itself is never algebraic here. Indeed if $s \neq 0$ were algebraic then $\cosh s$ would be transcendental by the Lindemann–Weierstrass theorem, contradicting Theorem 3.2. This is consistent with, and a special case of, the theorem of Calcut [3] that a hyperbolic triangle with all angles rational multiples of π has transcendental sides.

5 The system at $\cosh s = \varphi$

We now fix the system of Table 1 with $\cosh s = \varphi$ and study the polygons occurring at integer multiples of its base edge. Following the catalogue’s notation we write a letter for the multiplier and a number for the number of sides, so that A3 is a triangle of side s , B4 a square of side $2s$, G3 a triangle of side $7s$, and D ∞ an apeirogon of side $4s$.

Lemma 5.1. *The base edge satisfies $\cosh s = \varphi$, and it is the edge of the regular tiling $\{5, 4\}$, and of $\{5, 6\}$, $\{3, 10\}$ and $\{10, 5\}$ at twice its length. Moreover*

$$\cosh \frac{s}{2} = \frac{\varphi}{\sqrt{2}}, \quad \cosh s = \varphi, \quad \cosh 2s = \varphi^3, \quad \cosh \frac{7s}{2} = \frac{\varphi^7}{\sqrt{2}}.$$

Proof. For $\{5, 4\}$, Lemma 2.1 with $n = 5$ and $\theta = \pi/2$ gives $\cosh(s/2) = \cos(\pi/5)/\sin(\pi/4) = \varphi/\sqrt{2}$, whence $\cosh s = 2\varphi^2/2 - 1 = \varphi^2 - 1 = \varphi$. For $\{5, 6\}$ one finds $\cosh(s'/2) = \cos(\pi/5)/\sin(\pi/6) = \varphi = \cosh s$, so $s' = 2s$; the computation for $\{3, 10\}$ and $\{10, 5\}$ is identical, with $\cos(\pi/3)/\sin(\pi/10) = \varphi$ and $\cos(\pi/10)/\sin(\pi/5) = \varphi$. Finally $\cosh(ks) =$

symbol	n	edge	interior angle	$\cos \theta$
A3	3	s	a	$1/\varphi$
A4	4	s	$\pi - 2a$	$1/\varphi^3 = \sqrt{5} - 2$
A5	5	s	$\pi/2$	0
A10	10	s	t	$-1/\varphi^2$
A20	20	s	$\pi + a - t$	root of $x^4 + 8x^3 + 4x^2 - 8x - 4$
B3	3	$2s$	$\pi/5$	$(1 + \sqrt{5})/4$
B4	4	$2s$	a	$1/\varphi$
B5	5	$2s$	$\pi/3$	$1/2$
B10	10	$2s$	$2\pi/5$	$(\sqrt{5} - 1)/4$
B ∞	∞	$2s$	$\pi - 2a$	$1/\varphi^3$
D ∞	∞	$4s$	$4a - \pi$	$8\sqrt{5} - 17$
G3	3	$7s$	$7a - 2\pi$	$1 - \varphi^{-14}$

Table 2: The twelve polygons of the system, and their angles in the lattice.

$T_k(\cosh s) = T_k(\varphi)$ for the Chebyshev polynomial T_k , and $2 \cosh^2(ks/2) = T_k(\varphi) + 1$; for $k = 7$ one computes $T_7(\varphi) + 1 = 377\varphi + 233 = \varphi^{14}$, using $F_{14} = 377$ and $F_{13} = 233$. $\square$

Remark 5.2. Three regular tilings therefore share the edge $2s$: $\{5, 6\}$, $\{3, 10\}$ and $\{10, 5\}$, that is the vertex combinations (5^6) , (3^{10}) and (10^5) . This is the only coincidence of three regular tilings at one edge known to us, and Table 2 exhibits it as the three polygons B5, B3, B10 whose angles are $\pi/3$, $\pi/5$ and $2\pi/5$.

Remark 5.3. The full symmetry group of $\{5, 4\}$ is the triangle group $(2, 4, 5)$, which by Takeuchi's classification [10] is arithmetic with invariant trace field $\mathbb{Q}(\sqrt{5})$. That is the structural reason the quantities $\cosh(ks)$ are algebraic integers of $\mathbb{Z}[\varphi]$.

Theorem 5.4. *Put $a = \arccos(1/\varphi)$ and $t = \arccos(-1/\varphi^2)$. The twelve polygons listed in Table 2 have the interior angles shown there. In particular every one of them lies in the additive group $\mathbb{Q}\pi + \mathbb{Z}a + \mathbb{Z}t$.*

Proof of the first two rows, the rest being similar. By Lemma 2.1 and Lemma 5.1, $\sin(\theta_3/2) = \frac{1}{2} \cdot \sqrt{2}/\varphi = 1/(\sqrt{2}\varphi)$ and, using $2\varphi + 1 = \varphi^3$, $\cos(\theta_3/2) = \sqrt{\varphi^3}/(\sqrt{2}\varphi) = \sqrt{\varphi}/\sqrt{2}$. Hence $\cos \theta_3 = 1 - 2\sin^2(\theta_3/2) = 1 - 1/\varphi^2 = 1/\varphi$, which is the assertion $\theta_3 = a$. Likewise $\sin(\theta_4/2) = 1/\varphi$ and $\cos(\theta_4/2) = 1/\sqrt{\varphi}$, so with $\zeta_3 = e^{i\theta_3/2} = (\varphi^{3/2} + i)/(\sqrt{2}\varphi)$ and $\zeta_4 = e^{i\theta_4/2} = (\sqrt{\varphi} + i)/\varphi$ we get

$$\zeta_3^2 \zeta_4 = \frac{1 + i\sqrt{\varphi}}{\varphi} \cdot \frac{\sqrt{\varphi} + i}{\varphi} = \frac{i(1 + \varphi)}{\varphi^2} = i,$$

that is $\theta_3 + \theta_4/2 = \pi/2$, or $2\theta_3 + \theta_4 = \pi$. $\square$

Corollary 5.5. $\cos G3 = L_7/\varphi^7 = 1 - \varphi^{-14} = (377\sqrt{5} - 841)/2$, where $L_7 = 29$ is a Lucas number.

Proof. Writing $\omega = e^{ia} = (1 + i\sqrt{\varphi})/\varphi$, which has modulus one because $1 + \varphi = \varphi^2$, one has $\cos(ka) = \operatorname{Re}(1 + i\sqrt{\varphi})^k/\varphi^k$. Expanding, $\operatorname{Re}(1 + i\sqrt{\varphi})^7 = 1 - 21\varphi + 35\varphi^2 - 7\varphi^3 = 29$, using $\varphi^2 = \varphi + 1$ and $\varphi^3 = 2\varphi + 1$. Since $G3 = 7a - 2\pi$ we get $\cos G3 = \cos 7a = 29/\varphi^7$, and $29 = L_7 = \varphi^7 - \varphi^{-7}$ gives the stated forms. $\square$

Computational observation 5.6. Writing $\operatorname{Re}(1 + i\sqrt{\varphi})^k = p_k + q_k\varphi$ with $p_k, q_k \in \mathbb{Z}$, the recursion $p_k = 2p_{k-1} - (p_{k-2} + q_{k-2})$, $q_k = 2q_{k-1} - (p_{k-2} + 2q_{k-2})$ gives $q_k = 0$ for $k \leq 20000$ only at $k = 1$ and $k = 7$. These are exactly the two multiples of the base edge at which the system carries an equilateral triangle.

Computational observation 5.7. The integer relation algorithm PSLQ, applied at 200 digits to (π, a, t) , finds no relation with coefficients below 10^{15} . We therefore treat the lattice of Theorem 5.4 as having rank three. The margin matters for what follows only through coefficients of size at most about 900, since a vertex combination is bounded by 2π divided by the smallest angle.

5.1 Enumerating the vertex combinations

Granting Observation 5.7, closing a vertex becomes linear. Writing $m(\cdot)$ for multiplicities and clearing denominators, the three coordinates of $\sum m(P)\theta_P = 2\pi$ in the basis (t, a, π) read

$$m(\text{A10}) - m(\text{A20}) = 0, \quad (4)$$

$$m(\text{A3}) + m(\text{B4}) + m(\text{A20}) - 2m(\text{A4}) - 2m(\text{B}\infty) + 4m(\text{D}\infty) + 7m(\text{G3}) = 0, \quad (5)$$

$$\begin{aligned} 30m(\text{A4}) + 15m(\text{A5}) + 30m(\text{A20}) + 6m(\text{B3}) + 10m(\text{B5}) \\ + 12m(\text{B10}) + 30m(\text{B}\infty) - 30m(\text{D}\infty) - 60m(\text{G3}) = 60. \end{aligned} \quad (6)$$

Computational observation 5.8. The system (4)–(6) has exactly 106 solutions in non-negative integers; these fall into 30 classes if polygons are identified by angle rather than by shape. The analogous system with right-hand side π , describing vertices lying in the interior of a longer edge, has exactly 14 solutions in 7 angle classes. Each of the twelve polygons occurs in at least one solution.

The count 106 agrees with the catalogue. Two structural consequences are worth recording, both immediate from the equations.

Proposition 5.9. *In any solution of (4)–(6), $m(\text{A10}) = m(\text{A20})$. Consequently a 20-gon and a decagon occur at exactly the same vertices, and in equal numbers.*

Proof. Immediate from (4). □

Proposition 5.10. *In any solution, $m(\text{G3}) \leq 1$.*

Proof. By (5), $7m(\text{G3}) \leq 2(m(\text{A4}) + m(\text{B}\infty))$. The polygons A4 and B ∞ both have interior angle $\pi - 2a$, and $5(\pi - 2a) > 2\pi$ numerically, so at most four of them meet at a vertex. Hence $7m(\text{G3}) \leq 8$ and $m(\text{G3}) \leq 1$. □

Proposition 5.10 is sharp in an instructive way: the bound is $8/7$, so a single unit of slack separates this system from one in which two large triangles could meet. In the systems of Section 7 with $r \in \{2, 3, 4\}$ the corresponding coefficient is $r < 8/2$ and repetitions do occur.

6 Augmentation, and when it applies

Two of the coincidences in Table 2, namely $\text{B4} = \text{A3}$ and $\text{B}\infty = \text{A4}$, deserve separate comment because only the first has a geometric explanation.

Theorem 6.1 (Augmentation). *Let $s > 0$ and $3 \leq m \leq \infty$, and suppose $\theta_m(s) + 2\theta_3(s) = \pi$. Then $\theta_m(2s) = \theta_3(s)$.*

Proof. Write $c = \cosh(s/2)$ and $u = \sin(\theta_3/2) = 1/(2c)$ by Lemma 2.1. The hypothesis gives $\sin(\theta_m/2) = \cos \theta_3 = 1 - 2u^2$, hence $\cos(\pi/m) = c(1 - 2u^2) = (1 - 2u^2)/(2u)$. At the doubled edge $\cosh(s) = 2c^2 - 1 = 1/(2u^2) - 1$, so

$$\sin \frac{\theta_m(2s)}{2} = \frac{\cos(\pi/m)}{\cosh s} = \frac{1 - 2u^2}{2u} \cdot \frac{2u^2}{1 - 2u^2} = u = \sin \frac{\theta_3(s)}{2}. \quad \square$$

Geometrically: attach a triangle of side s to each side of the m -gon. At each original vertex the angle becomes $\theta_m + 2\theta_3 = \pi$, so those vertices straighten away, and what remains is an m -gon whose sides consist of two base edges each and whose angles are θ_3 . The hypothesis is exactly the statement that a vertex of type $(3^4.m^2)$ closes.

Theorem 6.2 (Tripling). *Let $s > 0$ and $2 \leq n \leq \infty$, and suppose $\theta_3(s) + \theta_{2n}(s) = \pi$. Then $\theta_n(3s) = \theta_3(s)$.*

Proof. Write c for the parameter of Remark 2.2 and $u = \sin(\theta_3/2) = 1/(2c)$. The hypothesis gives $\sin(\theta_{2n}/2) = \cos(\theta_3/2)$, hence $\cos(\pi/2n) = c\sqrt{1 - u^2}$ and

$$\cos(\pi/n) = 2\cos^2(\pi/2n) - 1 = 2c^2\left(1 - \frac{1}{4c^2}\right) - 1 = \frac{4c^2 - 3}{2}.$$

Since the parameter at the tripled edge is $T_3(c) = 4c^3 - 3c = c(4c^2 - 3)$,

$$\sin \frac{\theta_n(3s)}{2} = \frac{\cos(\pi/n)}{T_3(c)} = \frac{4c^2 - 3}{2c(4c^2 - 3)} = \frac{1}{2c} = u. \quad \square$$

Geometrically the hypothesis says that a vertex of type $(3.2n.3.2n)$ closes. Attach a triangle of side s to every second side of the $2n$ -gon. Each of its $2n$ vertices then carries $\theta_{2n} + \theta_3 = \pi$ and straightens away, leaving an n -gon whose sides are three base edges long and whose angles are θ_3 . By Remark 2.2 the theorem is not confined to $\mathbb{H}^2$: at $n = 2$ it applies on the sphere to the cuboctahedron (3.4.3.4), where $c = \sqrt{3}/2$, the edge is 60° , and the conclusion degenerates correctly to $\cos(\pi/2) = 0$.

Theorem 6.3 (Doubling criterion). *For $3 \leq n, m \leq \infty$ and $s > 0$,*

$$\theta_m(2s) = \theta_n(s) \iff \frac{\cos(\pi/m)}{\cos(\pi/n)} = \frac{\cosh s}{\cosh(s/2)}.$$

Proof. By Lemma 2.1, $\theta_m(2s) = \theta_n(s)$ if and only if $\cos(\pi/m)/\cosh(s) = \cos(\pi/n)/\cosh(s/2)$. $\square$

Proposition 6.4. *In the system of Section 5 one has $\cosh s/\cosh(s/2) = \sqrt{2}$, and the equation $\cos(\pi/m) = \sqrt{2}\cos(\pi/n)$ has exactly two solutions with $3 \leq n, m \leq \infty$, namely $(n, m) = (3, 4)$ and $(n, m) = (4, \infty)$. These give B4 = A3 and B ∞ = A4 respectively.*

Proof. $\cosh s/\cosh(s/2) = \varphi/(\varphi/\sqrt{2}) = \sqrt{2}$. Since $\cos(\pi/m) \leq 1$ we need $\cos(\pi/n) \leq 1/\sqrt{2}$, hence $\pi/n \geq \pi/4$ and $n \leq 4$. For $n = 3$, $\cos(\pi/m) = \sqrt{2}/2$ forces $m = 4$; for $n = 4$, $\cos(\pi/m) = 1$ forces $m = \infty$. $\square$

Proposition 6.5. *The hypothesis of Theorem 6.1, with a general n -gon in place of the triangle, is equivalent to the criterion of Theorem 6.3 only when $n = 3$.*

Proof. Writing $x = \cos(\pi/n)$ and $c = \cosh(s/2)$, the augmentation hypothesis $\theta_m(s) + 2\theta_n(s) = \pi$ gives $\cos(\pi/m) = c - 2x^2/c$, while the criterion gives $\cos(\pi/m) = x(2c - 1/c)$. Their difference, multiplied by c , factors as $(1 - 2x)(c^2 + x)$. The second factor is positive, so the two agree if and only if $x = 1/2$, that is $n = 3$. $\square$

Thus B4 = A3 arises from the augmentation picture, while B ∞ = A4 satisfies the criterion without any such construction behind it.

7 Triangles of proportional area

The system of Section 5 contains a triangle at seven times the base edge whose area is also exactly seven times that of the base triangle. We classify this phenomenon.

Proposition 7.1. *Let $s > 0$, let $r \geq 2$ be an integer, and write $\theta(s)$ for the interior angle of the equilateral triangle of side s . The following are equivalent:*

1. $\text{area}(\triangle(rs)) = r \cdot \text{area}(\triangle(s));$
2. $\theta(rs) = r\theta(s) - (r-1)\pi/3.$

Moreover for each r there is exactly one such s .

Proof. The area of a hyperbolic triangle with angles θ is $\pi - 3\theta$, so (1) reads $\pi - 3\theta(rs) = r(\pi - 3\theta(s))$, which rearranges to (2). For uniqueness, put $f(s) = \text{area}(\triangle(rs))/\text{area}(\triangle(s))$. As $s \rightarrow 0$ the geometry is asymptotically Euclidean and areas scale as the square of the side, so $f(s) \rightarrow r^2$; as $s \rightarrow \infty$ both areas tend to π and $f(s) \rightarrow 1$. The function f is continuous, and computation shows it to be strictly decreasing, so it attains the value $r \in (1, r^2)$ exactly once. $\square$

Condition (2) is again a relation between half-angles at two edge multiples, and the elimination of Theorem 3.2 applies verbatim. Writing $\zeta = e^{i\theta(s)/2} = (u + \frac{i}{2})/c$ and $\zeta_r = e^{i\theta(rs)/2} = (u_r + \frac{i}{2})/T_r(c)$ with $c = \cosh(s/2)$, $u = \sqrt{c^2 - \frac{1}{4}}$ and $u_r = \sqrt{T_r(c)^2 - \frac{1}{4}}$, condition (2) says

$$\zeta^r = e^{i(r-1)\pi/6} \zeta_r, \quad (7)$$

whose imaginary part, after eliminating u and u_r by resultants, is a polynomial in c that is even, hence a polynomial in $\cosh s = 2c^2 - 1$.

Theorem 7.2. *The edge of Proposition 7.1 satisfies, for $r = 2, 3, 4, 7$ respectively,*

$$\cosh s = 1 + \sqrt[3]{2} + \sqrt[3]{4} = \frac{1}{\sqrt[3]{2} - 1}, \quad \frac{5 + \sqrt{33}}{4}, \quad \frac{3 + \sqrt{33}}{4}, \quad \varphi,$$

with respective minimal polynomials over $\mathbb{Q}$

$$y^3 - 3y^2 - 3y - 1, \quad 2y^2 - 5y - 1, \quad 2y^2 - 3y - 3, \quad y^2 - y - 1.$$

In particular $\cosh s_3 - \cosh s_4 = \frac{1}{2}$, and indeed $y \mapsto y + \frac{1}{2}$ carries $2y^2 - 5y - 1$ to $2y^2 - 3y - 3$.

Proof. Carry out the elimination described above for (7), factor the resulting polynomial in $\cosh s$ over $\mathbb{Q}$, and select the irreducible factor vanishing at the numerical value of $\cosh s$; the four factors are those listed. Solving each gives the stated closed forms. For $r = 2$ the cubic $y^3 - 3y^2 - 3y - 1$ is irreducible over $\mathbb{Q}$; substituting $y = 1 + z + z^2$ and reducing modulo $z^3 = 2$ gives 0, so $1 + \sqrt[3]{2} + \sqrt[3]{4}$ is its real root, and $1 + z + z^2 = (z^3 - 1)/(z - 1) = 1/(\sqrt[3]{2} - 1)$. $\square$

The case $r = 7$ is the system of Section 5: the triangle G3 has $\theta = 7a - 2\pi$ and therefore area $7(\pi - 3a)$.

The polygons themselves are listed in Table 3; the system $r = 7$ is Table 2. In each case the base tiling recorded in the catalogue appears among the vertex combinations: $(A3^6.A6^2)$ at $r = 2$, $(C3^6.C6^{18})$ at $r = 3$ and $(A3^4.A6^2)$ at $r = 4$.

	$r = 2$	$r = 3$	$r = 4$
cosh s	$1 + \sqrt[3]{2} + \sqrt[3]{4}$	$(5 + \sqrt{33})/4$	$(3 + \sqrt{33})/4$
rank	2	3	4
basis	π, a	π, a, g	π, a, b, d
	$A3 = a$	$A3 = a$	$A3 = a$
	$A6 = \pi - 3a$	$A6 = g$	$A6 = \pi - 2a$
	$B3 = 2a - \pi/3$	$B3 = \pi - 2g$	$B6 = a$
	$B\infty = \pi - 4a$	$C3 = 3a - 2\pi/3$	$B\infty = b$
		$C6 = \pi/3 - a$	$D3 = 4a - \pi$
			$D\infty = d$
			$F\infty = \pi - 3b - d$
vertices	16	12	14
flat	6	5	5

Table 3: The three smaller systems of Proposition 7.1, reduced. The letter records the edge multiple, so C6 is a hexagon of side $3s$. Here $a = \theta_3(s)$ throughout, while g, b, d are the extra generators carried respectively by A6, B ∞ and D ∞ .

Computational observation 7.3. Reducing each of these systems as in Section 5, and scanning all $n \leq 300$ and all edge multiples $k \leq 15$, one finds: for $r = 2, 3, 4, 7$ the systems contain 4, 5, 7, 12 polygons respectively, in lattices of rank 2, 3, 4, 3, with 16, 12, 14, 106 vertex combinations; whereas for $2 \leq r \leq 15$ with $r \notin \{2, 3, 4, 7\}$ the only polygons in the span are the two triangles defining the system, and no vertex closes. The four rich values are exactly those for which $\cosh s$ has degree at most three over $\mathbb{Q}$.

Remark 7.4. The four fields are $\mathbb{Q}(\sqrt[3]{2})$, $\mathbb{Q}(\sqrt{33})$, $\mathbb{Q}(\sqrt{33})$ and $\mathbb{Q}(\sqrt{5})$. Only $r = 3$ and $r = 4$ share one, which is what forces their difference to be rational; see Section 8.

8 Limitations and open problems

Scaling the decision procedure

The elimination of Theorem 3.2 was carried out with nested radicals for the constants $\cos(\pi/n)$. This is adequate for small n but degenerates quickly. Of the four systems that resisted a first attempt, the two with moderate coefficient fields have since been completed and appear in Table 1: 1.1555, carrying pentagons and dodecagons, and 1.2537, which is the regular tiling $\{4, 5\}$ with decagons and 30-gons intruding. The two that remain are 1.0377, whose combination (3.4.8.40) carries a 40-gon, and 1.0531, whose two combinations (3.5.6.18) and (3.6².9) carry an 18-gon together with a 9-gon. Working throughout in a single cyclotomic field $\mathbb{Q}(\zeta_N)$ with $N = \text{lcm}\{2n\}$, and using modular or evaluation-interpolation resultants, should remove the obstruction. The system at 1.0531 is the one of greatest interest, since its pentagons admit no symmetry and its smallest tilings are 16-uniform.

Completeness of the catalogue

The catalogue is believed complete, with the explicit caveat that nothing rules out a system hiding between two listed edges. Theorem 3.4 decides equality and inequality of two given edges; what is needed for completeness is a bound, in terms of the degree M of a vertex combination, on the height of P_m , together with an enumeration of combinations up to that degree. Both appear to be within reach.

Intrusion into a regular tiling

Two entries of Table 1 have the same shape: a regular tiling together with one further combination that brings in a pair of much larger polygons. At 1.0612 the tiling $\{5, 4\} = (5^4)$ is accompanied by (3.4.10.20), and at 1.2537 the tiling $\{4, 5\} = (4^5)$ is accompanied by $(4^2.10.30)$. Both intruding combinations contain a decagon, and in both the second large polygon is a multiple of it. The two regular tilings are dual to one another. Whether these are the first two members of a family, and what the rule for the second large polygon is, we do not know.

Finiteness of a scaled hybrid system

Observation 5.8 presupposes the list of twelve polygons. We verified by search that no other pair (n, k) with $n \leq 2000$, $k \leq 60$ has its angle in the rank-three lattice, but the argument is circular in the following sense: the lattice was read off from the twelve polygons. What is missing is an argument that no disjoint cluster of further polygons can close vertices among themselves at the same edge.

Rank as an invariant

In each system studied the polygons split into a part lying in $\mathbb{Q}\pi + \mathbb{Z}a$ and a small family carrying the remaining generators and one relation among them: $\{A_{10}, A_{20}\}$ with $A_{10} + A_{20} = \pi + A_3$ in the system of Section 5; the apeirogons $\{B_\infty, D_\infty, F_\infty\}$ with $3B_\infty + D_\infty + F_\infty = \pi$ in the system $r = 4$; and $\{A_6, B_3\}$ with $2A_6 + B_3 = \pi$ in $r = 3$. Whether this is a general phenomenon, and what governs the rank, we do not know.

The shared field at $r = 3$ and $r = 4$

The identity $\cosh s_3 - \cosh s_4 = \frac{1}{2}$ of Theorem 7.2 is a consequence of both edges lying in $\mathbb{Q}(\sqrt{33})$ with the same coefficient on $\sqrt{33}$; their difference is then forced to be rational. Why these two systems share a quadratic field, when no other pair in the family does, we cannot say.

Reproducibility

All computations are reproduced by scripts depending only on the `mpmath` and `sympy` libraries. They recompute every closed form, re-derive the enumerations both on the integer lattice and by direct verification of the angle sums at 120 decimal digits, and re-run the completeness scans; a pass or failure is reported for each individual claim in this paper.

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